AMD DEVELOPER HACKATHON · ACT III

OmNi-Ledger

Deterministic supply-chain control for AI-assisted pharmaceutical intake

AI may extract and propose.
Deterministic rules decide what is allowed.

PROPOSED TRACK · REINVENT COMMERCE

Solo-founder prototype · Egypt · Technical evaluation build



An alert is not a control.

Pharmaceutical supply chains need rules that remain effective after an invoice has been read and stock has entered the warehouse.

01 Wrong item or batch

A plausible document extraction can still refer to the wrong SKU, lot, expiry, or quantity. Human review alone can miss a mismatch.

IDENTITY + SCHEMA VALIDATION

02 Expired stock

If an expired batch becomes available inventory, the error can travel into allocation, picking, packing, and dispatch.

DETERMINISTIC HARD BLOCK

03 Recall after dispatch

A recall must identify remaining stock and affected recorded customer orders—not just change a red status badge.

BATCH TRACE + AUDIT

THE DESIGN QUESTION

How can AI help process invoices without becoming the authority that changes critical inventory?

The answer is a clear trust boundary between probabilistic extraction and deterministic execution.

02 / APPROACH

AI reads. Rules decide. Evidence proves.

One connected control path—from document fields to warehouse state transitions and verifiable records.

01 / EXTRACT

Gemma on AMD

Untrusted structured fields

02 / VALIDATE

Strict policy gate

Dates · SKU · quantity · authority

03 / CONTROL

Warehouse core

QC · FEFO · scan · recall

04 / PROVE

Audit evidence

SHA-256 chain · state seal · Merkle root

Core invariant

The model never writes inventory balances directly.

Live inference runs in the secure Node CLI. The public judge page runs deterministic local scenarios.

Operational rules—not UI promises.

The judge edition exposes controls implemented in the shared domain core. Each visible decision is reproducible in the local scenario.

Expiry & receiving gate

An expired batch is rejected before it can become available stock; posting follows the QC workflow.

NEGATIVE CONTROL

FEFO & scan verification

Allocation uses eligible stock; picking verifies both item identity and the allocated batch.

ORDER STATE MACHINE

Recall & quarantine

A documented recall requires separate initiator and approver identities, then moves available stock in the selected warehouse into quarantine.

RECALLED_BATCH GUARDS

Audit integrity

Ledger movements and audit events are hash-linked; state-seal verification can reveal inconsistent data.

TAMPER-EVIDENT · NOT A LEGAL SIGNATURE

The same core powers the interactive judge demo and the live-CLI post-extraction decision.

Batch recall: trace backward, control forward.

Fictional demonstration scenario — computed by the packaged domain core, not manually authored dashboard numbers.

SUPPLIER

United Pharmaceutical Supply Co.

100 boxes received

WAREHOUSE

WH-OCT6 · QUARANTINE

70 boxes held

CUSTOMER

Al-Shifa Pharmacy 6th of October

30 boxes dispatched

POST-RECALL PICK ATTEMPT

HARD BLOCK RECALLED_BATCH

The previously allocated open order cannot be picked.

RECALL REFERENCE

DEMO-RECALL-017

Reason + separated approval recorded

Scope: batch-level trace through recorded warehouse dispatch. No patient, downstream sub-pharmacy, or individual serial tracking is claimed.

Trust is a result you can inspect.

The report connects the business decision to the ledger, actor, time, and integrity chain.

LEDGER CHAIN

VALID

Movement records are linked through prior hashes.

AUDIT CHAIN

VALID

Control events preserve their own event sequence.

STATE SEAL

VERIFIED

State hash and Merkle root are checked against the sample state.

EXAMPLE EVIDENCE FIELDS

EVENT HASH	STATE HASH	MERKLE ROOT
sha256(previousHash + canonicalEvent)	derived from current state	tree root over event/state hashes

Integrity hashes are not encryption, not a legal signature, and not proof of production security by themselves.

Strong claims have explicit boundaries.

The prototype's value comes from testable behavior—not from claiming to solve every downstream traceability problem.

Proven in the packaged tests

Valid and expired receipt paths, recall/quarantine, post-recall pick block, recorded customer trace, transfer guard under recall, and ledger/audit verification.

LOCAL AUTOMATED TESTS

Live AMD runs on Token Factory

Owner-run CLI requests with Gemma: valid invoice ACCEPTED, expired invoice BLOCKED (EXPIRED_GOODS). IDs in LIVE_RUN_EVIDENCE.md.

OWNER-CAPTURED EVIDENCE

Outside current scope

Patient-level and downstream sub-pharmacy tracing; individual GS1 serialization; customer returns; multi-user production concurrency.

NOT IMPLEMENTED

Not claimed

Regulatory certification, zero hallucinations, unhackable storage, production readiness, or guaranteed competition results.

NO OVERCLAIMS

The live request runs in Node/CLI; fixture mode never claims to be live inference.

AI-assisted operations.
Human accountability.
Verifiable control.

OmNi-Ledger was developed by a solo founder in Egypt through a phone-first, resource-constrained workflow. The mission is to make critical workflows safer to inspect and harder to bypass—not to hand operational authority to a language model.

LET THE EVIDENCE SPEAK

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